

printing pattern in python

```
In [3]: print('# # # #')
        print('# # # #')
        print('# # # #')
        print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [4]: for i in range(1,5):
        i=i+1
        print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [5]: for i in range (1,5):
        if i<=5:
            print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [6]: for j in range(4):
        print('#')
```

```
#
#
#
#
```

```
In [7]: for j in range(4):
        print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [8]: for i in range(4):
        print('#',end = "")
```

```
####
```

```
In [11]: for j in range(4):
          print('#', end=" ")

          for i in range(4):
```

```
print('#', end=" ")
```

```
# # # # # # # # # # # # # # # # # #
```

```
In [12]: for j in range(4):
          print('#', end=" ")

          print()

          for j in range(4):
              print('#', end=" ")
```

```
# # # #
# # # #
```

```
In [13]: for j in range(4):
          print('#', end=" ")

          print()

          for j in range(4):
              print('#', end=" ")

          print()

          for j in range(4):
              print('#', end=" ")

          print()

          for j in range(4):
              print('#', end=" ")
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [14]: for i in range(4):
          for j in range(4):
              print('#', end=" ")
          print()
          # please use debug mode in pycharm
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [15]: for i in range(4):
          for j in range(i+1):
              print('#', end=" ")
          print()
```

```
#
# #
# # #
# # # #
```

```
In [16]: for i in range(1,5):
          print("# "*i)
```

```
#
# #
# # #
# # # #
```

```
In [18]: for i in range(1,5):
          for j in range(4):
              if i>j:
                  print("#",end=" ")
          print()
```

```
#
# #
# # #
# # # #
```

```
In [19]: # right angle triangle pattern
          for i in range(1,6):
              print(' '*i)
```

```
*
* *
* * *
* * * *
* * * * *
```

```
In [21]: # inverted right angle triangle pattern
          for i in range(5,0,-1):
              print(' '*i)
```

```
* * * * *
* * * *
* * *
* *
*
```

```
In [23]: # pyramid pattern
          for i in range(1,6):
              print('*(5-i)+' * '*(2*i-1))
```

```
*
* * *
* * * * *
* * * * * * *
* * * * * * * *
```

```
In [25]: # inverted pyramid pattern
          for i in range(5,0,-1):
              print('*(5-i)+' * '*(2*i-1))
```

```

* * * * *
* * * * *
* * * * *
* * *
*

```

```

In [29]: # diamond pattern
for i in range(1,6):
    print('*(5-i)+' * '(2*i-1))
for i in range(4,0,-1):
    print('*(5-i)+' * '(2*i-1))

```

```

*
* * *
* * * * *
* * * * * * *
* * * * * * * *
* * * * * *
* * * *
* *
*

```

```

In [41]: # hallow square pattern
for i in range(5):
    for j in range(5):
        if i==0 or i==4 or j==0 or j==4:
            print('*',end='')
        else:
            print(' ',end='')
    print()

```

```

*****
*   *
*   *
*   *
*****

```

```

In [42]: # full square pattern
for i in range(5):
    print('* * * * *')

```

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

```

```

In [44]: # right angle triangle(number pattern)
for i in range(1,6):
    print(' '.join(str(x) for x in range(1, i + 1)))

```

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

```
In [49]: # inverted right angle triangle(number pattern)
for i in range(5,0,-1):
    print(' '.join(str(x) for x in range(1,i+1)))
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
In [62]: # floyd's tringle
num=1
for i in range(1,6):
    for j in range(1,i+1):
        print(num,end=' ')
        num+=1
    print()
```

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

```
In [2]: # hallow right angle triangle
for i in range(1,6):
    for j in range(1,i+1):
        if j==1 or j==i or i==5:
            print('*',end=' ')
        else:
            print(' ',end=' ')
    print()
```

```
*
* *
*  *
*   *
* * * * *
```

```
In [19]: # hallow pyramid pattern
for i in range(1, 6):
    for j in range(5 - i):
        print(' ',end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2 or i == 5:
            print('*', end=' ')
        else:
            print(' ',end=' ')
    print()
```

```
      *
     * *
    *   *
   *     *
  *       *
 * * * * *
* * * * * * * *
```

```
In [12]: # hallow diamond pattern
n = 5
for i in range(1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print('*', end=' ')
        else:
            print(' ', end=' ')
    print()

for i in range(n - 1, 0, -1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print('*', end=' ')
        else:
            print(' ', end=' ')
    print()
```

```

      *
    * *
  *   *
 *     *
*       *
 *     *
  *   *
    * *
      *

```

```
In [14]: # hallow diamond (number pattern)
n = 5
for i in range(1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print(i, end=' ')
        else:
            print(' ', end=' ')
    print()

for i in range(n - 1, 0, -1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print(i, end=' ')
        else:
            print(' ', end=' ')
    print()
```

```

      1
    2 2
  3   3
4    4
5    5
  4   4
    3 3
      2 2
        1

```

```

In [20]: # butterfly pattern
n = 5
for i in range (1, n + 1):
    for j in range(1, i + 1):
        print(j, end=' ')
    for j in range(2 * (n - i)):
        print(' ',end=' ')
    for j in range(1, i + 1):
        print(j, end=' ')
    print()

for i in range (n, 0, -1):
    for j in range(1, i + 1):
        print(j, end=' ')
    for j in range(2 * (n - i)):
        print(' ',end=' ')
    for j in range(1, i + 1):
        print(j, end=' ')
    print()

```

```

1           1
1 2         1 2
1 2 3       1 2 3
1 2 3 4     1 2 3 4
1 2 3 4 5 1 2 3 4 5
1 2 3 4 5 1 2 3 4 5
1 2 3 4     1 2 3 4
1 2 3       1 2 3
1 2         1 2
1           1

```

```

In [21]: n = 5
for i in range (1, n + 1):
    for j in range(1, i + 1):
        print('*', end=' ')
    for j in range(2 * (n - i)):
        print(' ',end=' ')
    for j in range(1, i + 1):
        print('*', end=' ')
    print()

for i in range (n, 0, -1):
    for j in range(1, i + 1):
        print('*', end=' ')
    for j in range(2 * (n - i)):
        print(' ',end=' ')

```

```

for j in range(1, i + 1):
    print('*', end=' ')
print()

```

```

*               *
* *           * *
* * *       * * *
* * * *   * * * *
* * * * * * * * *
* * * * * * * * *
* * * *   * * * *
* * *       * * *
* *           * *
*               *

```

```

In [25]: # hallow number pyramid
n = 5
for i in range (1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')

    for j in range(1, 2 * i):
        if j == 1 or j == 2 * i - 1 or i == n:
            print(i, end=' ')
        else:
            print(' ', end=' ')
    print()

```

```

      1
     2 2
    3   3
   4    4
  5 5 5 5 5 5 5

```

```

In [28]: # full pyramid
n = 5
for i in range (1, n + 1):

    for j in range(n - i):
        print(' ', end=' ')

    for j in range(2 * i - 1):
        print('*', end=' ')

    print()

```

```

      *
     * * *
    * * * * *
   * * * * * *
  * * * * * * *
 * * * * * * * *

```

```

In [30]: # inverted full star pyramid
n = 5
for i in range (n, 0, -1):

    for j in range(n - i):

```



```

        print(' ',end=' ')

    for j in range(2 * i - 1):
        print('*', end=' ')

    print()

```

```

* * * * *
* * * * *
* * * *
* * *
*

```

In [31]: *# Left aligned pyramid pattern*

```

n = 5
for i in range (1, n + 1):

    for j in range(i):
        print('*',end=' ')
    print()

n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=' ')

    print()

```

```

*
* *
* * *
* * * *
* * * * *
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

In [32]: *# right alined pyramid pattern*

```

n = 5
for i in range (1, n + 1):

    for j in range(n - i):
        print(' ',end=' ')

    for j in range(1, i + 1):
        print(j,end=' ')

    print()

n = 5

for i in range(1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')

```

```
for j in range(i):  
    print('*', end=' ')  
  
print()
```

```
    1  
  1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5  
      *  
    * *  
  * * *  
* * * *  
* * * * *
```

In []: